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CODATA

Fundamental Physical Constants

Version 2.5.2

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Chapter 1

Introduction

1.1 DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA.
<https://www.nist.gov/programs-projects/codata-values-fundamental-physical-constants>.

- o **Online documentation** <https://milanskocic.github.io/codata/>
- o **PDF documentation** <https://github.com/MilanSkocic/codata/tree/main/doc/latex/pdf>
- o **Man pages** <https://github.com/MilanSkocic/codata/tree/main/doc/manpages>
- o **Examples** <https://github.com/MilanSkocic/codata/tree/main/example>
- o **Source code** <https://github.com/MilanSkocic/codata>
- o **Python wrapper** <https://pypi.org/project/pycodata>
- o **Binaries** <https://github.com/MilanSkocic/codata/releases>

It provides:

- codata constants 2022
- codata constants 2018
- codata constants 2014
- codata constants 2010

The codata constant from 2022 were integrated in the Fortran stdlib. See <https://github.com/fortran-lang/stdlib/releases/tag/v0.7.0>.

1.2 NOTES

If you need older values for the CODATA constants you can use 'codata' otherwise use the standard library.

If you only need sources for the codata constants that you can integrate directly in your sources you may be interested by this repo https://github.com/vmagnin/fundamental_constants.

Chapter 2

Installation

2.1 FROM BINARIES

Download the binaries for your operating system from <https://github.com/MilanSkocic/codata/releases>. The archives contains 3 folders bin, lib and share that you can simply copy/paste somewhere on your PATH (usually \$HOME/.local). For Windows, you can add .local folder to your %USERPROFILE% and add it to your PATH.

```
$ unzip codata-linux-x86_64-2.5.2.zip codata
$ cp -v codata/* your/path/
```

2.2 FROM SOURCES

You need to install a compilation toolchain for Fortran. The Fortran compiler must support the f2018 standard.

You also need to install fpm (Fortran Package Manager) which can be downloaded at <https://github.com/fortran-lang/fpm/releases/>.

A Makefile allows to compile the static and the shared library. It uses fpm for compiling the dependencies and the source code.

```
$ ./configure
$ make
$ make install
```

2.3 PYTHON WRAPPER

If you also need to build the Python wrapper. There is script build.sh which builds the static and shared library and also the Python wrapper.

You need to install a compilation toolchain for C. The C compiler must support the c11 standard.

```
$ ./build.sh
```


Chapter 3

codata(1)

3.1 NAME

codata - Command line for codata

3.2 SYNOPSIS

codata [OPTIONS] [REGEX_PATTERN ...]

3.3 DESCRIPTION

codata is a command line interface which prints all the codata constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

3.4 OPTIONS

`-year, -y YEAR` Year of the codata constants: 2022, 2018, 2014, 2010

`-pattern, -p PATTERN` Regex pattern for filtering the constants.

`-value, -a` Show only the value.

`-error, -e` Show only the uncertainty.

`-usage` Show usage text and exit.

`-help` Show help text and exit.

`-verbose` Display additional information.

`-version` Show version information and exit.

3.5 NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

3.6 EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p 'molar.*gas','electron.*eV'
codata '[B,b]oltzmann.*eV'
```

3.7 SEE ALSO

`codata(3)`

Chapter 4

codata(3)

4.1 NAME

codata - library for fundamental physical constants

4.2 LIBRARY

codata (-libcodata, -lcodata)

4.3 SYNOPSIS

use codata include "codata.h" import pycodata

4.4 DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA <https://www.nist.gov/programs-projects/codata-values-fundamental-physical-constants>. A C API allows usage from C, or can be used as a basis for other wrappers. A python wrapper allows easy usage from Python.

The latest codata constants 2022 <https://pml.nist.gov/cuu/Constants> were integrated in stdlib <https://github.com/fortran-lang/stdlib/releases/tag/> since version 0.7.0. The constants are implemented as derived type which carries the name, the value, the uncertainty and the unit. This library is complementary to the constants defined in the stdlib by providing older values for the constants. The latest values (2022) do not have the year as a suffix in their name. Older values (2010, 2014, 2018) can be used and they feature the year as a suffix in their name.

All codata (physical) constants are defined as a derived type `codata_constant_type`. All the codata constants are provided as double precision reals. The names are quite long and can be aliased with shorter names. The derived type `codata_constant_type` defines 4 members and 2 procedures.

```
type :: codata_constant_type
  !! Derived type for representing a Codata constant.
  character(len=64) :: name
  real(dp) :: value
  real(dp) :: uncertainty
  character(len=32) :: unit
contains
  procedure :: print
  procedure :: to_real_sp
```

```

    procedure :: to_real_dp
    generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type

```

A module level interface to `_real` is available for getting the constant value or uncertainty of a constant.

The C API exposes a structure `codata_constant_type` that defines the same members as in Fortran.

```

type, bind(C) :: capi_constant_type
    !! Derived type for representing a Codata constant in C.
    character(kind=c_char) :: name(65)
    real(c_double) :: value
    real(c_double) :: uncertainty
    character(kind=c_char) :: unit(33)
end type capi_constant_type

typedef struct codata_constant_type{
    char name[65];
    double value;
    double uncertainty;
    char unit[33];
}cct;

```

The Python wrapper encapsulates the members in a dictionary with the keys being `name`, `value`, `uncertainty` and `unit`.

4.5 NOTES

To use `codata` within your `fpm` <https://github.com/fortran-lang/fpm> project, add the following lines to your file:

```

[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }

```

4.6 EXAMPLE

Example in Fortran

```

program example_in_f
use codata
implicit none
print '(A)', '##### EXAMPLE IN FORTRAN #####'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty

```

```

print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

```

Example in C

```

#include <stdio.h>
#include "codata.h"
int main(void){
printf("##### EXAMPLE IN C #####\n");
printf("%s\n", "# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n", "# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n", "# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n", "# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}

```

Example in Python

```

import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])

```

4.7 SEE ALSO

gsl(3), codata(1)

4.8 CODATA 2022

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY

- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90

- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM
- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO

- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT

- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT

- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220

- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV

- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C

- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS

- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT
- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT

- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY
- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV
- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY

- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

4.9 CODATA 2018

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018
- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018

- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018
- ATOMIC_UNIT_OF_PERMITTIVITY_2018
- ATOMIC_UNIT_OF_TIME_2018
- ATOMIC_UNIT_OF_VELOCITY_2018
- AVOGADRO_CONSTANT_2018
- BOHR_MAGNETON_2018
- BOHR_MAGNETON_IN_EV_T_2018
- BOHR_MAGNETON_IN_HZ_T_2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- BOHR_MAGNETON_IN_K_T_2018
- BOHR_RADIUS_2018
- BOLTZMANN_CONSTANT_2018
- BOLTZMANN_CONSTANT_IN_EV_K_2018
- BOLTZMANN_CONSTANT_IN_HZ_K_2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018
- CLASSICAL_ELECTRON_RADIUS_2018
- COMPTON_WAVELENGTH_2018
- CONDUCTANCE_QUANTUM_2018

- CONVENTIONAL_VALUE_OF_AMPERE_90_2018
- CONVENTIONAL_VALUE_OF_COULOMB_90_2018
- CONVENTIONAL_VALUE_OF_FARAD_90_2018
- CONVENTIONAL_VALUE_OF_HENRY_90_2018
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_OHM_90_2018
- CONVENTIONAL_VALUE_OF_VOLT_90_2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_WATT_90_2018
- COPPER_X_UNIT_2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2018
- DEUTERON_ELECTRON_MASS_RATIO_2018
- DEUTERON_G_FACTOR_2018
- DEUTERON_MAG_MOM_2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- DEUTERON_MASS_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- DEUTERON_MASS_IN_U_2018
- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018

- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018
- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018
- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018
- ELEMENTARY_CHARGE_2018
- ELEMENTARY_CHARGE_OVER_H_BAR_2018

- FARADAY_CONSTANT_2018
- FERMI_COUPLING_CONSTANT_2018
- FINE_STRUCTURE_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018
- HARTREE_ENERGY_2018
- HARTREE_ENERGY_IN_EV_2018
- HARTREE_HERTZ_RELATIONSHIP_2018
- HARTREE_INVERSE_METER_RELATIONSHIP_2018
- HARTREE_JOULE_RELATIONSHIP_2018
- HARTREE_KELVIN_RELATIONSHIP_2018
- HARTREE_KILOGRAM_RELATIONSHIP_2018
- HELION_ELECTRON_MASS_RATIO_2018
- HELION_G_FACTOR_2018
- HELION_MAG_MOM_2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- HELION_MASS_2018
- HELION_MASS_ENERGY_EQUIVALENT_2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- HELION_MASS_IN_U_2018
- HELION_MOLAR_MASS_2018
- HELION_PROTON_MASS_RATIO_2018
- HELION_RELATIVE_ATOMIC_MASS_2018
- HELION_SHIELDING_SHIFT_2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018
- HERTZ_HARTREE_RELATIONSHIP_2018
- HERTZ_INVERSE_METER_RELATIONSHIP_2018
- HERTZ_JOULE_RELATIONSHIP_2018

- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018
- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018
- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018
- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018
- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018

- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018
- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018
- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018
- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018
- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018

- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018
- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018
- NEUTRON_ELECTRON_MASS_RATIO_2018
- NEUTRON_G_FACTOR_2018
- NEUTRON_GYROMAG_RATIO_2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- NEUTRON_MAG_MOM_2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- NEUTRON_MASS_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_MASS_IN_U_2018
- NEUTRON_MOLAR_MASS_2018
- NEUTRON_MUON_MASS_RATIO_2018
- NEUTRON_PROTON_MAG_MOM_RATIO_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018
- NEUTRON_PROTON_MASS_RATIO_2018
- NEUTRON_RELATIVE_ATOMIC_MASS_2018

- NEUTRON_TAU_MASS_RATIO_2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018
- NUCLEAR_MAGNETON_2018
- NUCLEAR_MAGNETON_IN_EV_T_2018
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- NUCLEAR_MAGNETON_IN_K_T_2018
- NUCLEAR_MAGNETON_IN_MHZ_T_2018
- PLANCK_CONSTANT_2018
- PLANCK_CONSTANT_IN_EV_HZ_2018
- PLANCK_LENGTH_2018
- PLANCK_MASS_2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018
- PLANCK_TEMPERATURE_2018
- PLANCK_TIME_2018
- PROTON_CHARGE_TO_MASS_QUOTIENT_2018
- PROTON_COMPTON_WAVELENGTH_2018
- PROTON_ELECTRON_MASS_RATIO_2018
- PROTON_G_FACTOR_2018
- PROTON_GYROMAG_RATIO_2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- PROTON_MAG_MOM_2018
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- PROTON_MAG_SHIELDING_CORRECTION_2018
- PROTON_MASS_2018
- PROTON_MASS_ENERGY_EQUIVALENT_2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- PROTON_MASS_IN_U_2018
- PROTON_MOLAR_MASS_2018
- PROTON_MUON_MASS_RATIO_2018

- PROTON_NEUTRON_MAG_MOM_RATIO_2018
- PROTON_NEUTRON_MASS_RATIO_2018
- PROTON_RELATIVE_ATOMIC_MASS_2018
- PROTON_RMS_CHARGE_RADIUS_2018
- PROTON_TAU_MASS_RATIO_2018
- QUANTUM_OF_CIRCULATION_2018
- QUANTUM_OF_CIRCULATION_TIMES_2_2018
- REDUCED_COMPTON_WAVELENGTH_2018
- REDUCED_MUON_COMPTON_WAVELENGTH_2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018
- REDUCED_PLANCK_CONSTANT_2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S_2018
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018
- REDUCED_PROTON_COMPTON_WAVELENGTH_2018
- REDUCED_TAU_COMPTON_WAVELENGTH_2018
- RYDBERG_CONSTANT_2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2018
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018
- SECOND_RADIATION_CONSTANT_2018
- SHIELDED_HELION_GYROMAG_RATIO_2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018

- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018
- SPEED_OF_LIGHT_IN_VACUUM_2018
- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
- TAU_COMPTON_WAVELENGTH_2018
- TAU_ELECTRON_MASS_RATIO_2018
- TAU_ENERGY_EQUIVALENT_2018
- TAU_MASS_2018
- TAU_MASS_ENERGY_EQUIVALENT_2018
- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
- TAU_MUON_MASS_RATIO_2018
- TAU_NEUTRON_MASS_RATIO_2018
- TAU_PROTON_MASS_RATIO_2018
- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018
- TRITON_PROTON_MASS_RATIO_2018
- TRITON_RELATIVE_ATOMIC_MASS_2018

- TRITON_TO_PROTON_MAG_MOM_RATIO_2018
- UNIFIED_ATOMIC_MASS_UNIT_2018
- VACUUM_ELECTRIC_PERMITTIVITY_2018
- VACUUM_MAG_PERMEABILITY_2018
- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018
- W_TO_Z_MASS_RATIO_2018

4.10 CODATA 2014

- LATTICE_SPACING_OF_SILICON_2014
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2014
- ALPHA_PARTICLE_MASS_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ALPHA_PARTICLE_MASS_IN_U_2014
- ALPHA_PARTICLE_MOLAR_MASS_2014
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2014
- ANGSTROM_STAR_2014
- ATOMIC_MASS_CONSTANT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2014
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2014
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2014
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- ATOMIC_UNIT_OF_ACTION_2014
- ATOMIC_UNIT_OF_CHARGE_2014
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2014
- ATOMIC_UNIT_OF_CURRENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014
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- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014
- ATOMIC_UNIT_OF_ENERGY_2014
- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
- ATOMIC_UNIT_OF_MOMUM_2014
- ATOMIC_UNIT_OF_PERMITTIVITY_2014
- ATOMIC_UNIT_OF_TIME_2014
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- AVOGADRO_CONSTANT_2014
- BOHR_MAGNETON_2014
- BOHR_MAGNETON_IN_EV_T_2014
- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014

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- COMPTON_WAVELENGTH_OVER_2_PI_2014
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- CU_X_UNIT_2014
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
- DEUTERON_ELECTRON_MASS_RATIO_2014
- DEUTERON_G_FACTOR_2014
- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- DEUTERON_MASS_IN_U_2014
- DEUTERON_MOLAR_MASS_2014
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
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- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
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- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014
- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014
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- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2014
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- ELECTRON_VOLT_JOULE_RELATIONSHIP_2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2014
- ELEMENTARY_CHARGE_2014
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- FARADAY_CONSTANT_2014
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- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2014
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- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2014
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- HARTREE_ENERGY_IN_EV_2014
- HARTREE_HERTZ_RELATIONSHIP_2014
- HARTREE_INVERSE_METER_RELATIONSHIP_2014
- HARTREE_JOULE_RELATIONSHIP_2014
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- HELION_MASS_2014
- HELION_MASS_ENERGY_EQUIVALENT_2014
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- HELION_MASS_IN_U_2014
- HELION_MOLAR_MASS_2014
- HELION_PROTON_MASS_RATIO_2014
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- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014
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- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014

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- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
- INVERSE_METER_JOULE_RELATIONSHIP_2014
- INVERSE_METER_KELVIN_RELATIONSHIP_2014
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- JOULE_HERTZ_RELATIONSHIP_2014
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- KELVIN_HARTREE_RELATIONSHIP_2014
- KELVIN_HERTZ_RELATIONSHIP_2014
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- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014
- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
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- KILOGRAM_KELVIN_RELATIONSHIP_2014
- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014

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- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
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- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
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- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
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- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014
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- NEUTRON_G_FACTOR_2014
- NEUTRON_GYROMAG_RATIO_2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014
- NEUTRON_MAG_MOM_2014
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- NEUTRON_MASS_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_2014
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- NEUTRON_MASS_IN_U_2014
- NEUTRON_MOLAR_MASS_2014
- NEUTRON_MUON_MASS_RATIO_2014
- NEUTRON_PROTON_MAG_MOM_RATIO_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014
- NEUTRON_PROTON_MASS_RATIO_2014
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- PLANCK_CONSTANT_2014
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- PROTON_G_FACTOR_2014
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- PROTON_MAG_MOM_2014
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- PROTON_MOLAR_MASS_2014
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- PROTON_NEUTRON_MAG_MOM_RATIO_2014
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- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
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- SHIELDED_HELION_GYROMAG_RATIO_2014
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_HELION_MAG_MOM_2014
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_2014
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- SHIELDED_PROTON_MAG_MOM_2014
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- STEFAN_BOLTZMANN_CONSTANT_2014
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- TAU_MASS_2014
- TAU_MASS_ENERGY_EQUIVALENT_2014
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- TAU_MASS_IN_U_2014
- TAU_MOLAR_MASS_2014
- TAU_MUON_MASS_RATIO_2014
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- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010
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- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
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- CLASSICAL_ELECTRON_RADIUS_2010
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- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010
- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
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- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- ELECTRON_MASS_IN_U_2010
- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010

- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
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- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
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- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010
- HARTREE_ENERGY_2010
- HARTREE_ENERGY_IN_EV_2010
- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010

- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010
- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
- HERTZ_INVERSE_METER_RELATIONSHIP_2010
- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
- INVERSE_METER_JOULE_RELATIONSHIP_2010
- INVERSE_METER_KELVIN_RELATIONSHIP_2010
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010

- JOULE_ELECTRON_VOLT_RELATIONSHIP_2010
- JOULE_HARTREE_RELATIONSHIP_2010
- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010
- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
- KELVIN_JOULE_RELATIONSHIP_2010
- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
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- MOLAR_VOLUME_OF_SILICON_2010
- MO_X_UNIT_2010
- MUON_COMPTON_WAVELENGTH_2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- MUON_ELECTRON_MASS_RATIO_2010
- MUON_G_FACTOR_2010
- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- MUON_MASS_2010
- MUON_MASS_ENERGY_EQUIVALENT_2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- MUON_MASS_IN_U_2010
- MUON_MOLAR_MASS_2010
- MUON_NEUTRON_MASS_RATIO_2010
- MUON_PROTON_MAG_MOM_RATIO_2010
- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
- NATURAL_UNIT_OF_ENERGY_2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2010
- NATURAL_UNIT_OF_LENGTH_2010
- NATURAL_UNIT_OF_MASS_2010
- NATURAL_UNIT_OF_MOMUM_2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2010
- NATURAL_UNIT_OF_TIME_2010
- NATURAL_UNIT_OF_VELOCITY_2010
- NEUTRON_COMPTON_WAVELENGTH_2010
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010

- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
- NUCLEAR_MAGNETON_2010
- NUCLEAR_MAGNETON_IN_EV_T_2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- NUCLEAR_MAGNETON_IN_K_T_2010
- NUCLEAR_MAGNETON_IN_MHZ_T_2010
- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010

- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010
- PLANCK_LENGTH_2010
- PLANCK_MASS_2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010
- PLANCK_TEMPERATURE_2010
- PLANCK_TIME_2010
- PROTON_CHARGE_TO_MASS_QUOTIENT_2010
- PROTON_COMPTON_WAVELENGTH_2010
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- PROTON_ELECTRON_MASS_RATIO_2010
- PROTON_G_FACTOR_2010
- PROTON_GYROMAG_RATIO_2010
- PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- PROTON_MAG_MOM_2010
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- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
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- SHIELDED_PROTON_GYROMAG_RATIO_2010
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- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
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- TAU_ELECTRON_MASS_RATIO_2010
- TAU_MASS_2010
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- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
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- TRITON_G_FACTOR_2010
- TRITON_MAG_MOM_2010
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- TRITON_MASS_2010
- TRITON_MASS_ENERGY_EQUIVALENT_2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TRITON_MASS_IN_U_2010
- TRITON_MOLAR_MASS_2010
- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010
- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

Appendix A

CHANGELOG

A.1 VERSION 2.5.2

- Maintenance release. No API break.
- Use `prep(1)` for auto-generating the version parameter.
- Add new API for getting the version number:
- `version()` instead of `get_version()`.
- `get_version()` is marked as deprecated, it will be removed in the next major release 3.0 when the new codata constants will be released i.e. 2026/2027.

A.2 VERSION 2.5.1

- Code refactoring and cleaning: trailing spaces, empty lines, formatted comments.
- Remove specific modules for C API: defined in the same module as Fortran equivalents.
- No API break.

A.3 VERSION 2.5.0

- Add a `response_file` (https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html) capability for the CLI.
- Add rst output for the man pages.
- Integration of the man pages in the online documentation generated with sphinx.

A.4 VERSION 2.4.1

- Maintenance release.
- Fix option formatting in help text and man page.
- Refactoring

A.5 VERSION 2.4.0

- Add tables for each codata year containing all constants.
- Add CLI `codata(1)` for searching constants.
- Refactoring code.

A.6 VERSION 2.3.1

- Refactoring the `configure.sh` script.
- Remove support for 3.14t. No official release on `python.org`.
- If binaries for Python 3.14t are needed you need to compile them by

yourself.

A.7 VERSION 2.3.0

- Remove support for Python 3.9 and add support for Python 3.14(t).

A.8 VERSION 2.2.0

- Switch to UCRT64 for Windows binaries.
- Switch to sphinx documentation using `fspixh` (<https://github.com/jalvesz/fspix>).
- Update references with publication for codata 2022.
- Update compilation flags for compatibility with `stdlib`.

A.9 VERSION 2.1.1

- No code change.
- Code refractoring and cleaning
- Update CI/CD workflows.

A.10 VERSION 2.1.0

- Roll back to C API in Fortran code: easier maintenance.
- Roll back to compiled C extension for python: easier maintenance.

A.11 VERSION 2.0.1

- Fix bug in version for Fortran code.

A.12 VERSION 2.0.0

- Drop compiled extensions for Python.
- Pure Python code for constants auto-generated as it is the case for the Fortran code.
- Pure C code for constants auto-generated as it is the case for the Fortran code.
- API break:
- No more C API in the Fortran code.
- Use the pure C code to build a C library.

A.13 VERSION 1.2.2

- Fix conflict that could occur with C API modules. Add prefix in module names.
- Cleanup and refactoring.
- Documentation update.

A.14 VERSION 1.2.1

- Refactoring
- Merge back C API and python wrapper.

A.15 VERSION 1.2.0

- Refactoring
- Documentation update.

A.16 VERSION 1.1.0

- C API and Python wrapper moved to their own repositories.
- API break: C API is no more provided by default. Use the optional C wrapper.
- Code cleanup
- Documentation update

A.17 VERSION 1.0.0

- Add codata values for 2010, 2014 and 2018.
- Code refactoring and code cleaning.
- Documentation update and switch to only FORD documentation.
- Rewrite code generators in python.
- Generate source code for stdlib.
- API break: constants are defined as DT like in stdlib.

A.18 VERSION 0.10.0

- Remove remove generation of the version module.
- Add tests using the test-drive framework.
- Explicit parameter constants for Fortran and protected constants for C API.
- Minor fixes in documentation.
- Code cleanup.
- Merge of all code for autogeneration in one file.

A.19 VERSION 0.9.0

- No API changes.
- Automatic generation of the version module.
- Generic Makefiles for automatic the building process of the library and the pywrapper.
- Add targets: build, build_debug, test, test_debug.
- Minor fixes in documentation.

A.20 VERSION 0.8.2

- No API changes.
- Improve Makefile for generating the source code at each compilation.
- Source generator rewritten in Fortran.
- Switch to pyproject.toml for the Python wrapper.
- Minor fixes in documentation.

A.21 VERSION 0.8.1

- Use shared library in python wrapper.
- Minor fixes in documentation.

A.22 VERSION 0.8.0

- Back to the approach with a library.
- Compatible with fpm.
- Configuration file for setting all the environmental variables.
- Global makefile for building a static library (through fpm) and a shared library.
- Automatic copy of the necessary sources for the python wrapper.
- Python wrapper built with the static library

- no dependency on a shared library.
- sources and static library embeded in the python wrapper.
- FORD for documenting the Fortran code.
- Integration of the FORD documentation into the main documentation with sphinx.

A.23 VERSION 0.7.1

- Minor fixes in generator code
- Add automatic copy of c sources for the python wrapper.

A.24 VERSION 0.7.0

- Migrate documentation from doxygen to sphinx+breath.
- Add YEAR constant indicating the year of the codata constants.
- Refactoring

A.25 VERSION 0.6.0

- Created documentation.
- Fixed missing uncertainties for Cpython.

A.26 VERSION 0.5.0

- Changed the complete approach by not generating a library but only source files for different languages.
- Available languages: Fortran, C, python, CPython

A.27 VERSION 0.4.0

- Bring back pywrapper in the codata repository to sync versions.
- Improvements of the documentation.

A.28 VERSION 0.3.0

- Only last codata constants.

A.29 VERSION 0.2.1

- Integration of Intel Fortran compiler and MSVC in cmake scripts.
- Add specifications and instructions for compiling on Windows

A.30 VERSION 0.2.0

- Bug fixes for the codata 2010.
- Bug fixes in the tests linked to the codata 2010.
- Add python wrapper for the number of constants method.

A.31 VERSION 0.1.0

Implementation of:

- the parser of the codata raw data
- the generator of the Fortran modules
- the C API and C header
- the python wrapper (will be moved to its repository next release).

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